

Documentation for Noah-MP SSM-DA Evaluation Framework (postproc_02)

This directory (/scripts/postproc/postproc_02) contains the analytical scripts to generate a set of scientific figures and statistics evaluating the impact of Soil Moisture Data Assimilation (SSM-DA) on the Noah-MP land surface model over the Sebou-Saïss basin, focusing on fluxes, vegetation dynamics, and droughts.

Below is the detailed description of each figure, its scientific objective, and the required datasets and their **current availability**.

Figure 1: Spatial Context of the Study Area

- **Objective:** Establish the spatial context by identifying regions where vegetation-soil-irrigation interactions are likely to influence water balances.
- **Data Used:**
 - **Land Cover:** MODIS MCD12Q1 (IGBP, 1km) - *Assumed Available (LIS Input)*
 - **Irrigation Intensity:** FAO GMIA (or GRIPC/GIAM) - *Available*
- **Status:** Ready for the 3-day test layout, though maps are static.

Figure 2: Spatial Impact of DA on Surface Fluxes

- **Objective:** Map differences (Δ and Δ Anomaly) between SSM-DA and OL for E, T, ET, GPP, and NPP.
- **Data Used:**
 - **Noah-MP Output:** Evap_tavg, TVeg_tavg, GPP_tavg, NPP_tavg from OL and DA.
 - **Validation:** FAO WaPOR (ET, E, T, NPP) and FLUXSAT (GPP).
- **Status:** **MISSING VALIDATION DATA.** WaPOR and FLUXSAT need to be downloaded to /data/observations/. The script is a skeleton.

Figure 3: Statistical Evaluation by Ecosystem Type

- **Objective:** Identify ecosystems where assimilation significantly improves or degrades model performance using correlation (R) boxplots.
- **Data Used:**
 - **Noah-MP Output:** E, T, ET, NPP, GPP.
 - **Land Cover:** MODIS MCD12Q1.
 - **Validation:** FAO WaPOR, FLUXSAT.
- **Status:** **MISSING VALIDATION DATA.** Script is a skeleton.

Figure 4: Simulated Vegetation Dynamics

- **Objective:** Compare monthly LAI time series (OL vs SSM-DA) to assess phenology and drought response improvements.
- **Data Used:**
 - **Noah-MP Output:** LAI_tavg.
- **Status:** Ready. Can run on LIS outputs (though requires longer runs than 3 days for meaningful seasonal cycles).

Figure 5: Drought Categorization

- **Objective:** Quantify the effect of assimilation on spatial drought detection.
- **Data Used:**
 - **Noah-MP Output:** Root zone soil moisture (SoilMoist_tavg for layers).
- **Status:** Ready. Computes percentiles based on OL climatology. Requires multi-year simulations to establish robust climatologies.

Figure 6: Spatial Response to an Extreme Event

- **Objective:** Evaluate the capability of SSM-DA to reproduce spatial impacts of water stress on vegetation during a major drought.
- **Data Used:**
 - **Noah-MP Output:** LAI_tavg anomalies.
 - **Validation:** MODIS LAI MCD15A2H v6.
- **Status:** **PARTIALLY READY.** MODIS LAI is available, but detecting a major drought event requires multi-year LIS simulations.

Supplementary Statistical Analysis

- **Objective:** Comprehensive statistical validation (R, Anomaly R, Bias, RMSE, ubRMSE, NSE, KGE) for SM, ET, T, E, GPP, NPP, LAI.
- **Data Used:** All aforementioned LIS outputs and validation datasets.
- **Status:** **MISSING VALIDATION DATA.** Script is a skeleton.

Summary Table

Figure	Message	Data used from simulation (variables) if any	Data used from observation (if any)
Figure 1	Establish spatial context (vegetation-soil-irrigation)	None	MODIS MCD12Q1 (Land Cover), FAO GMIA/GRIPC (Irrigation)
Figure 2	Map spatial impact of DA on surface fluxes (Δ and Δ Anomaly)	Evap_tavg, TVeg_tavg, GPP_tavg, NPP_tavg	FAO WaPOR (ET, E, T, NPP), FLUXSAT (GPP)
Figure 3	Identify ecosystems where DA improves/degrades performance	E, T, ET, NPP, GPP	MODIS MCD12Q1, FAO WaPOR, FLUXSAT
Figure 4	Compare LAI dynamics to assess phenology and drought response	LAI_tavg	MODIS LAI MCD15A2H v6
Figure 5	Quantify effect of assimilation on spatial drought detection	Root zone soil moisture (SoilMoist_tavg)	None
Figure 6	Evaluate spatial vegetation response to a major drought event	LAI_tavg anomalies	MODIS LAI MCD15A2H v6
Supp Stats	Comprehensive statistical validation (R, SM, ET, T, E, GPP, NPP, LAI)		All validation datasets

Figure

Message

RMSE, KGE, NSE,
etc.)

**Data used from
simulation (variables
if any**

**Data used from
observation (if any)**